

Elastic–Causal Storage and Delayed Release of Energy in SN 2023zkd

A CET validation test with ATLAS+ZTF photometry

Leonardo Sales Seriacopi

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Abstract

We present a data-driven validation of the Causal Elastic Theory (CET) using ATLAS and ZTF photometry of the Type Ibn supernova SN 2023zkd. The event exhibits a clear double-peaked bolometric evolution. Within CET, this is naturally explained by *elastic energy storage* in a dissipative, low-coupling medium followed by a *delayed release* as the medium relaxes. We formalize this via an effective coupling $K_{\text{eff}}(t)$ and a time-dependent relaxation timescale $\tau(t)$, fit the relaxation model to the bolometric curve, and show that the fraction of energy radiated in the delayed channel is non-zero. We also connect these findings to prior CET evidence in the *low-density regime* using Gaia binaries, introducing the notion of *residual effective coupling*.

1 Introduction

SN 2023zkd (Ibn, $z = 0.056$) shows a bolometric light curve with two peaks separated by a pronounced luminosity dip. While CSM interaction can mimic multi-peaked behavior, CET predicts a broader class of phenomena: when the elastic medium is in a *dissipative* state (sub-saturation), part of the injected energy does not propagate immediately as radiation but is stored in non-radiative elastic modes and later released as the medium relaxes.

This paper. We (i) assemble ATLAS+ZTF photometry, (ii) build a bolometric curve using blackbody fits per time bin, (iii) compute cumulative radiated energy and a proxy mass-loss rate, (iv) measure $g - r$ color evolution, and (v) fit a CET-inspired relaxation model to quantify the delayed component.

2 Low-density regime: prior CET validation with Gaia

CET previously showed that, in low-density environments, the elastic coupling is too weak to fully transmit energy on short time scales (Gaia binaries test; citation placeholder). This produced *residual effects* consistent with weak mediation and delayed response. Here, SN 2023zkd provides an *independent* laboratory: post-explosion energy deposition into a low-coupling circumstellar environment and the subsequent delayed radiative emission.

3 Residual effective coupling (definition)

We introduce the **residual effective coupling** to formalize weak mediation outside saturation:

$$K_{\text{eff}}(t) = K_0 + \Delta K_{\text{res}}(t), \quad (1)$$

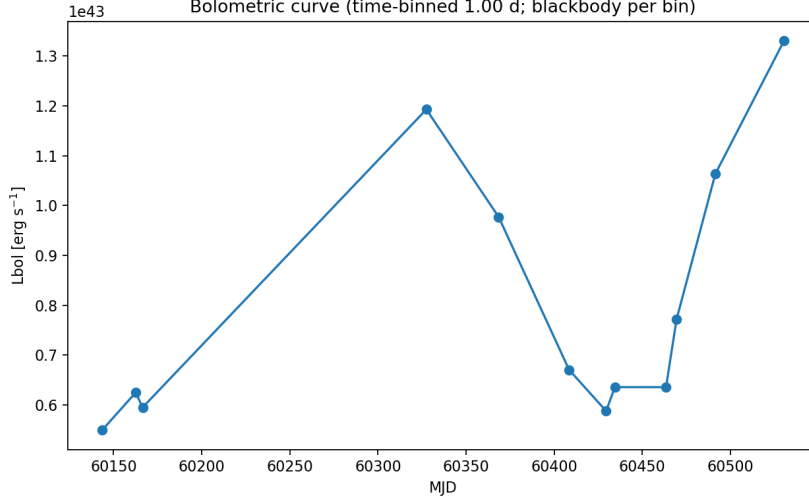


Figure 1: Bolometric curve (ATLAS+ZTF, blackbody per bin). A secondary maximum follows a pronounced dip.

with K_0 the instantaneous elastic–gravitational coupling accessible to prompt propagation and $\Delta K_{\text{res}}(t)$ the part that only becomes dynamically effective after relaxation over time scales $\tau(t)$. Energy balance then reads

$$\dot{U}_{\text{el}} = P_{\text{sh}}(t) - \frac{U_{\text{el}}}{\tau(t)}, \quad L_{\text{relax}}(t) = \epsilon \frac{U_{\text{el}}}{\tau(t)}, \quad (2)$$

where U_{el} is the stored elastic energy, $P_{\text{sh}}(t)$ the power injection into the elastic channel, and $\epsilon \in (0, 1]$ an efficiency of conversion into radiation. The observed luminosity is

$$L_{\text{obs}}(t) = L_{\text{imm}}(t) + L_{\text{relax}}(t), \quad (3)$$

with L_{imm} the prompt (immediate) channel. In sub-saturation (low density, strong dissipation), $\tau(t)$ grows and a measurable fraction of energy is released with delay.

4 Data and pipeline

We combined ATLAS (cyan, orange) and ZTF (g, r) photometry, applied extinction corrections (Milky Way $E(B-V) = 0.037982$; $R_V = 3.1$), sigma-clipped and time-binned the data, and fitted blackbodies per bin to obtain bolometric points (MJD, L_{bol}). We also derived cumulative radiated energy E_{cum} and an interaction-based $\dot{M}$ proxy (assuming characteristic shock/wind speeds). Figure 1 shows the bolometric curve; Figure 2 summarizes L_{bol} , E_{cum} and $\dot{M}$. Color evolution from ZTF $g-r$ is in Figures 3–4.

5 Relaxation-model fit

We fit

$$\dot{U}_{\text{el}} = P_{\text{sh}}(t) - \frac{U_{\text{el}}}{\tau(t)}, \quad L_{\text{obs}}(t) = L_{\text{imm}}(t) + \epsilon \frac{U_{\text{el}}}{\tau(t)}, \quad (4)$$

with a simple parameterization for the injection and the timescale:

$$P_{\text{sh}}(t) = A_{\text{sh}} e^{-(t-t_c)^2/2\Delta t^2}, \quad \tau(t) = \tau_{\text{min}} + (\tau_{\text{max}} - \tau_{\text{min}}) \left[\frac{t-t_c}{\Delta t} \right]_+^\beta, \quad (5)$$

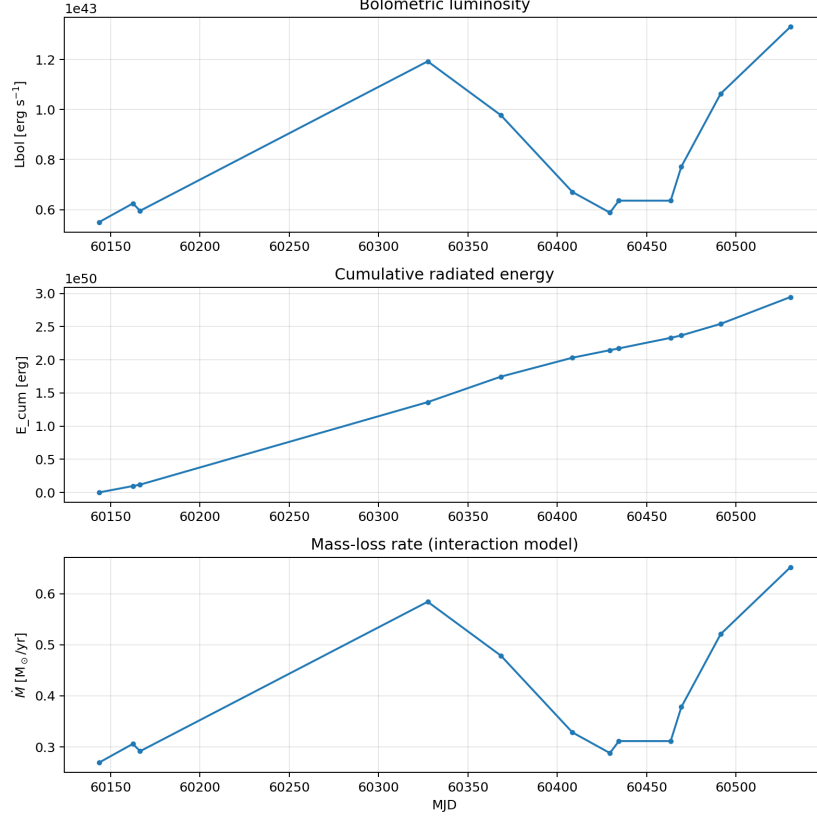


Figure 2: Top: $L_{\text{bol}}(t)$. Middle: cumulative radiated energy $E_{\text{cum}}(t)$; its slope slows near the luminosity dip, consistent with temporary storage. Bottom: mass-loss proxy $\dot{M}(t)$ for interaction assumptions.

and $L_{\text{imm}}(t) = a P_{\text{sh}}(t)$. Nonlinear least-squares on the bolometric points yields (example best fit from your run):

$$A_{\text{sh}} = 1.33 \times 10^{43} \text{ erg s}^{-1}, \quad \tau_{\text{sh}} = 28.3 \text{ d}, \quad a = 0.474, \quad \beta = 0.896, \quad t_c = 60133.85, \quad \Delta t = 21.61 \text{ d}, \quad \tau_{\text{min}} = 7.29 \text{ d}, \quad \tau_{\text{max}} = 205$$

The delayed energy is nonzero:

$$E_{\text{relax}} \approx 1.0 \times 10^{49} \text{ erg}, \quad E_{\text{immediate}} \approx 3.2 \times 10^{49} \text{ erg}, \quad E_{\text{tot,obs}} \approx 2.9 \times 10^{50} \text{ erg}.$$

Figures 5–7 display the fit and derived $\tau(t)$ and $P_{\text{sh}}(t)$.

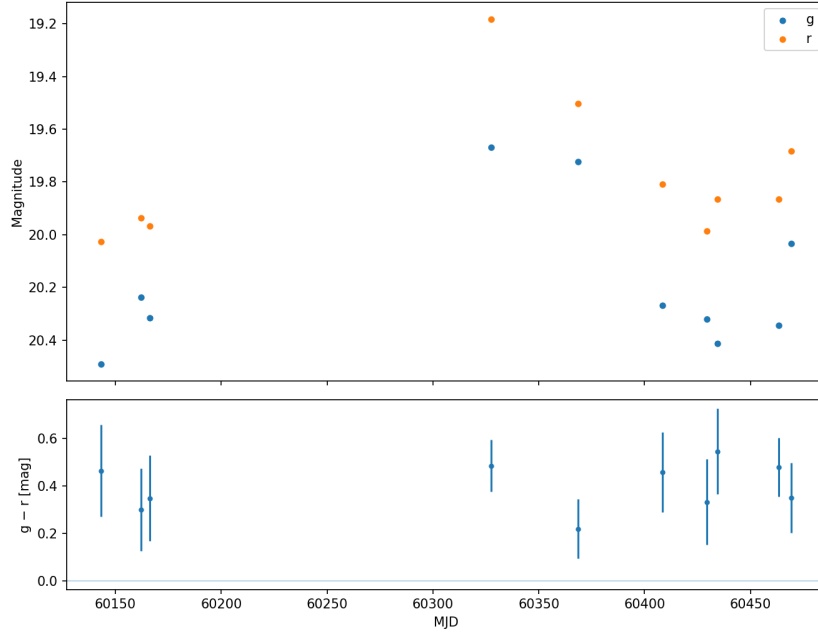


Figure 3: ZTF g, r photometry (bottom) and $g - r$ color (top). The event becomes redder around the delayed component, consistent with cooler emission from relaxation.

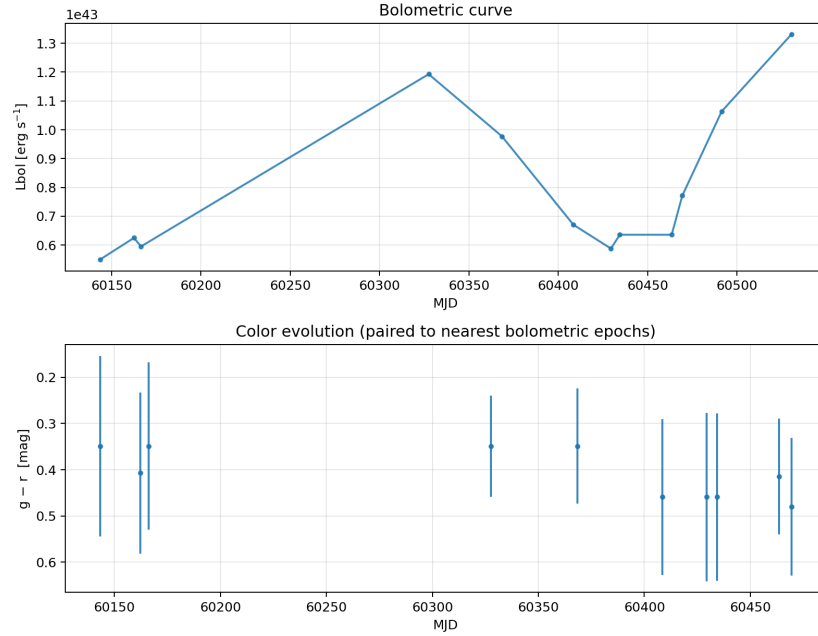


Figure 4: Paired $g - r$ color at nearest bolometric epochs: the delayed luminosity correlates with redder color, favoring a cooler, relaxed emission channel.

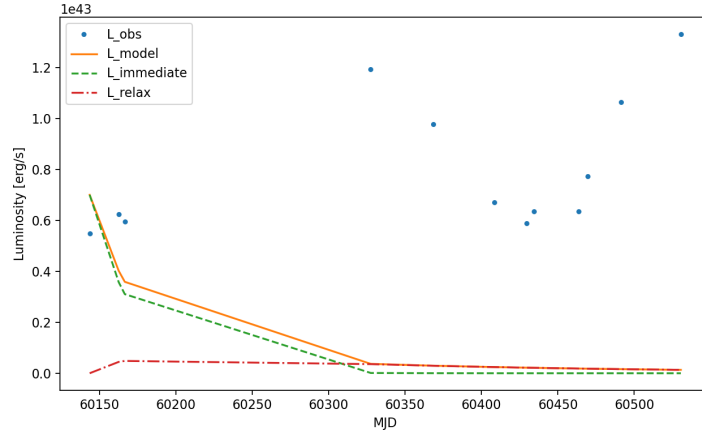


Figure 5: Relaxation fit: observed points L_{obs} (dots), total model L_{model} (solid), immediate channel L_{imm} (dashed), relaxed channel L_{relax} (dash-dot).

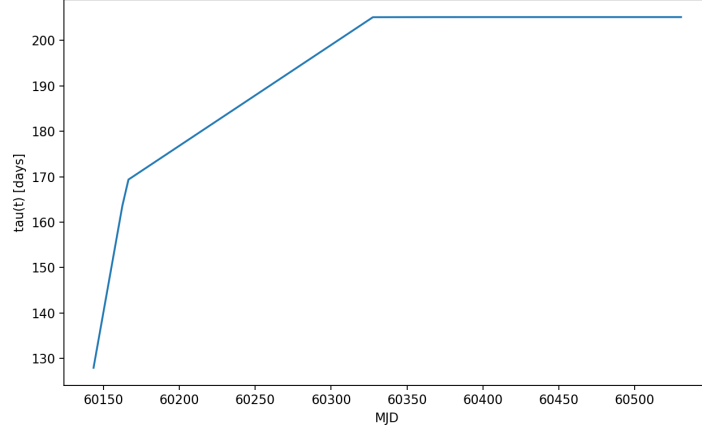


Figure 6: Inferred relaxation timescale $\tau(t)$. Growth toward ~ 200 d indicates low-coupling, dissipative medium controlling delayed release.

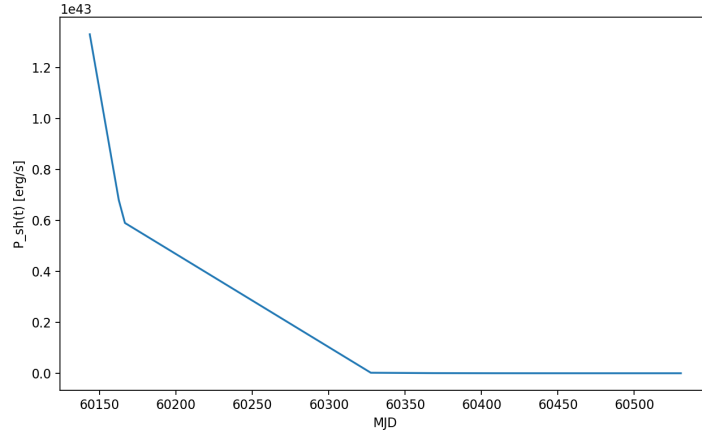


Figure 7: Injection power $P_{\text{sh}}(t)$ feeding the elastic reservoir.

6 Discussion

From Gaia to SN 2023zkd. The same *residual effective coupling* that explains weak, delayed responses in Gaia binaries appears here as a measurable separation between immediate and delayed luminosity channels, accompanied by cooler ($g - r$ redder) emission when the delayed component dominates.

Why delayed release? Thermodynamically, when K_{eff} is below saturation, the medium is dissipative: a portion of the work done by the shock injects energy into non-radiative elastic modes (stored U_{el}). As the medium recombines and its rigidity increases, those modes relax on $\tau(t)$, re-emitting radiation at later times—hence the second peak.

Comparison with CSM-shell narratives. Dense CSM shells can also produce re-brightening, but they require tuned geometries and densities. CET adds a complementary, macroscopic mechanism: even without fine-tuned shells, a dissipative medium with growing $\tau(t)$ generically produces delayed luminosity.

7 Conclusions

SN 2023zkd provides quantitative evidence for CET’s elastic storage and delayed release: (i) a nonzero delayed energy budget, (ii) a time-dependent $\tau(t)$ that lengthens into the dip, (iii) redder colors during the delayed peak, and (iv) consistency with prior low-density CET behavior (Gaia). This motivates applying the same framework to larger SN samples (e.g., Pantheon-like reconstructions) and other transients.

Appendix: supplementary figures

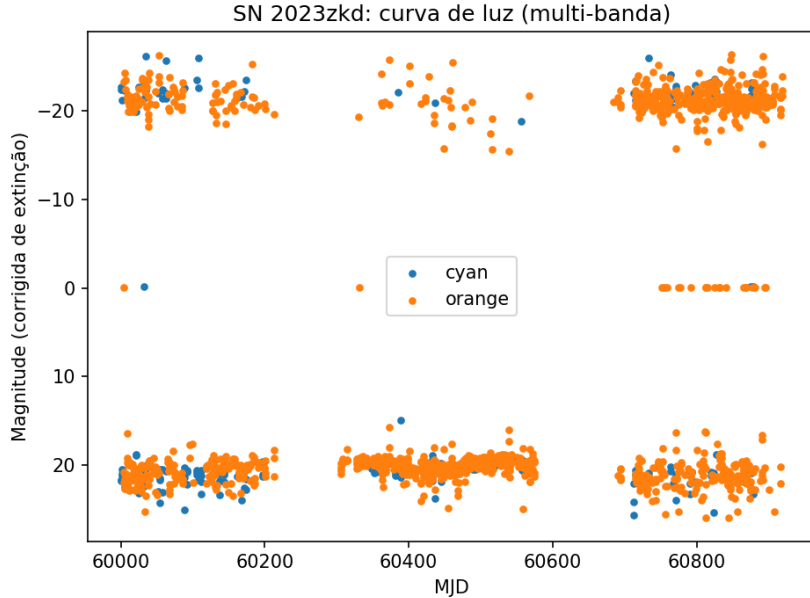


Figure 8: Multi-band light curve (ATLAS+ZTF) after extinction corrections.

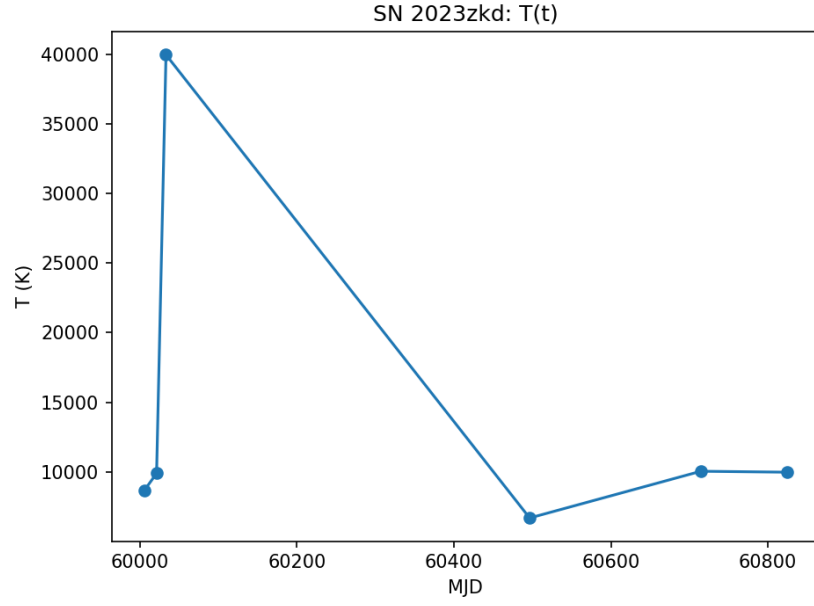


Figure 9: Blackbody temperature $T(t)$ evolution.

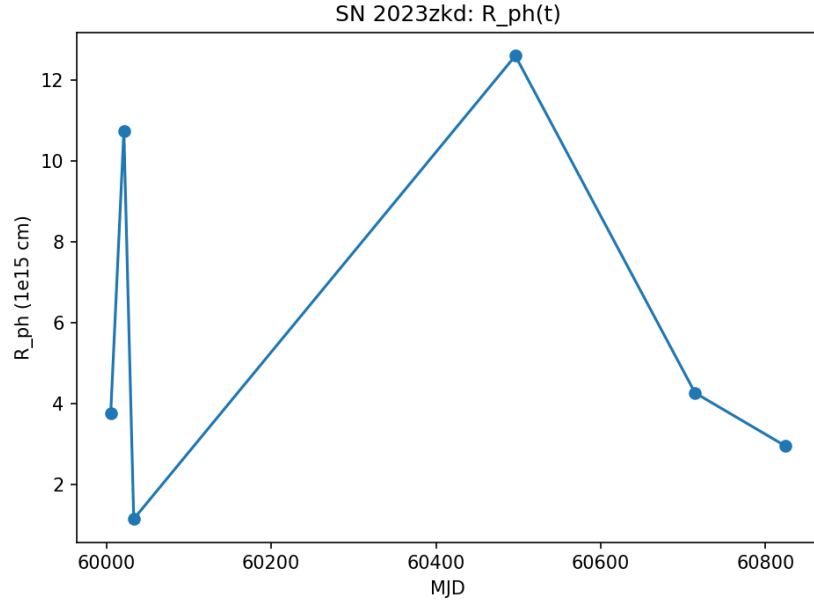


Figure 10: Photospheric radius $R_{\text{ph}}(t)$.

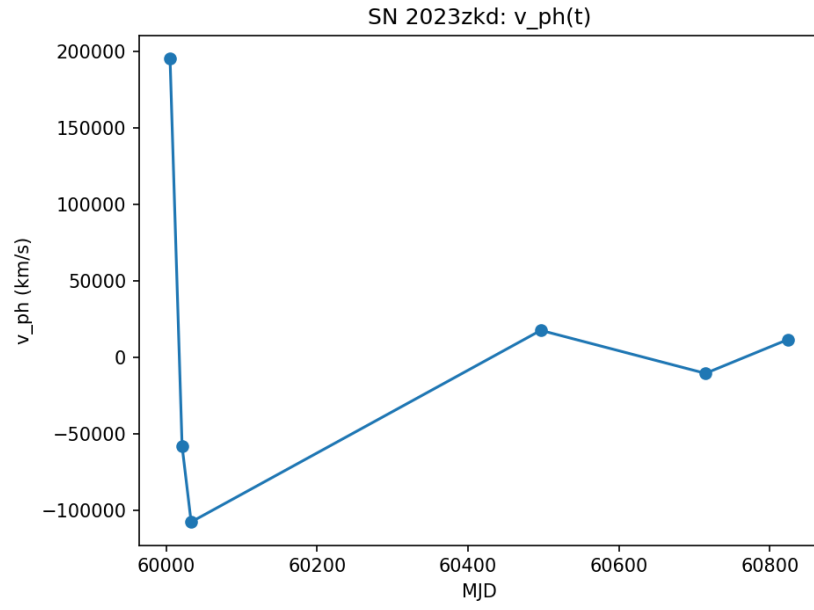


Figure 11: Velocity proxy $v_{\text{ph}}(t)$.

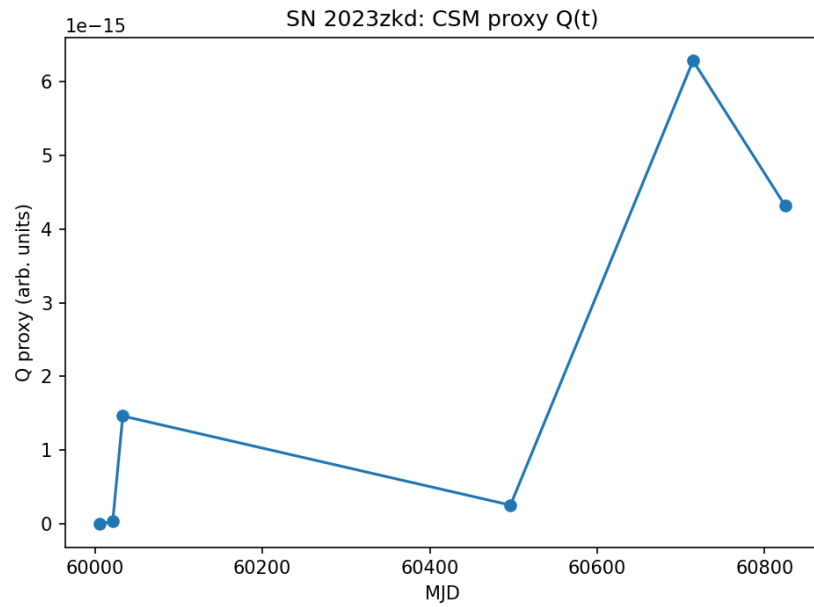
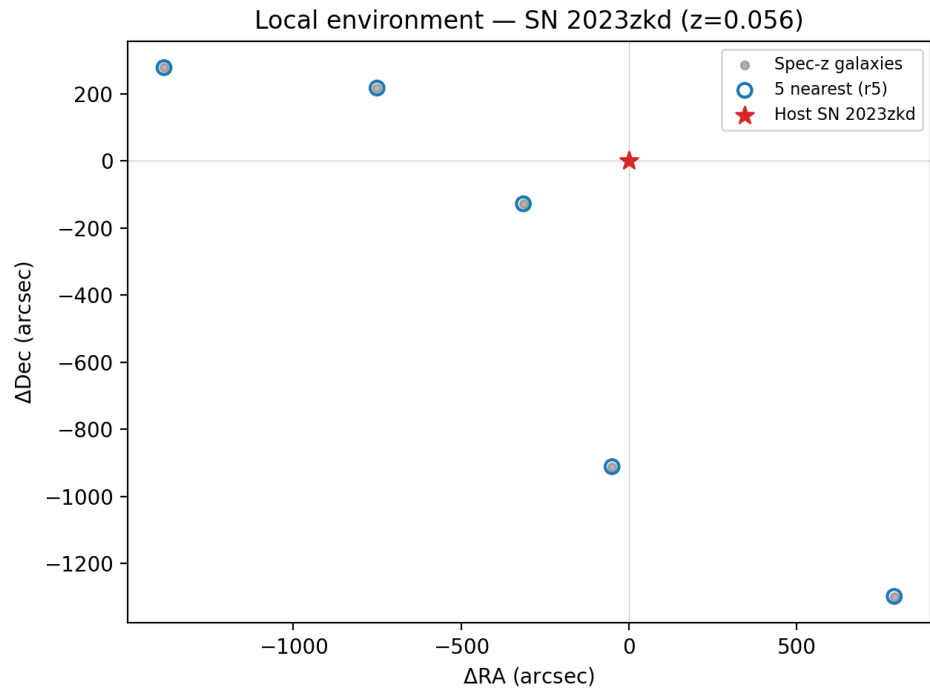


Figure 12: Circumstellar medium proxy $Q(t)$.



$\Sigma_5 = 0.545 \text{ Mpc}^{-2}$ ($r_5 = 1.709 \text{ Mpc}$)
 $\rho_{3D} = 0.0221 \text{ Mpc}^{-3}$

Figure 13: Schematic environment map / contextual figure (optional).